SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct
I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

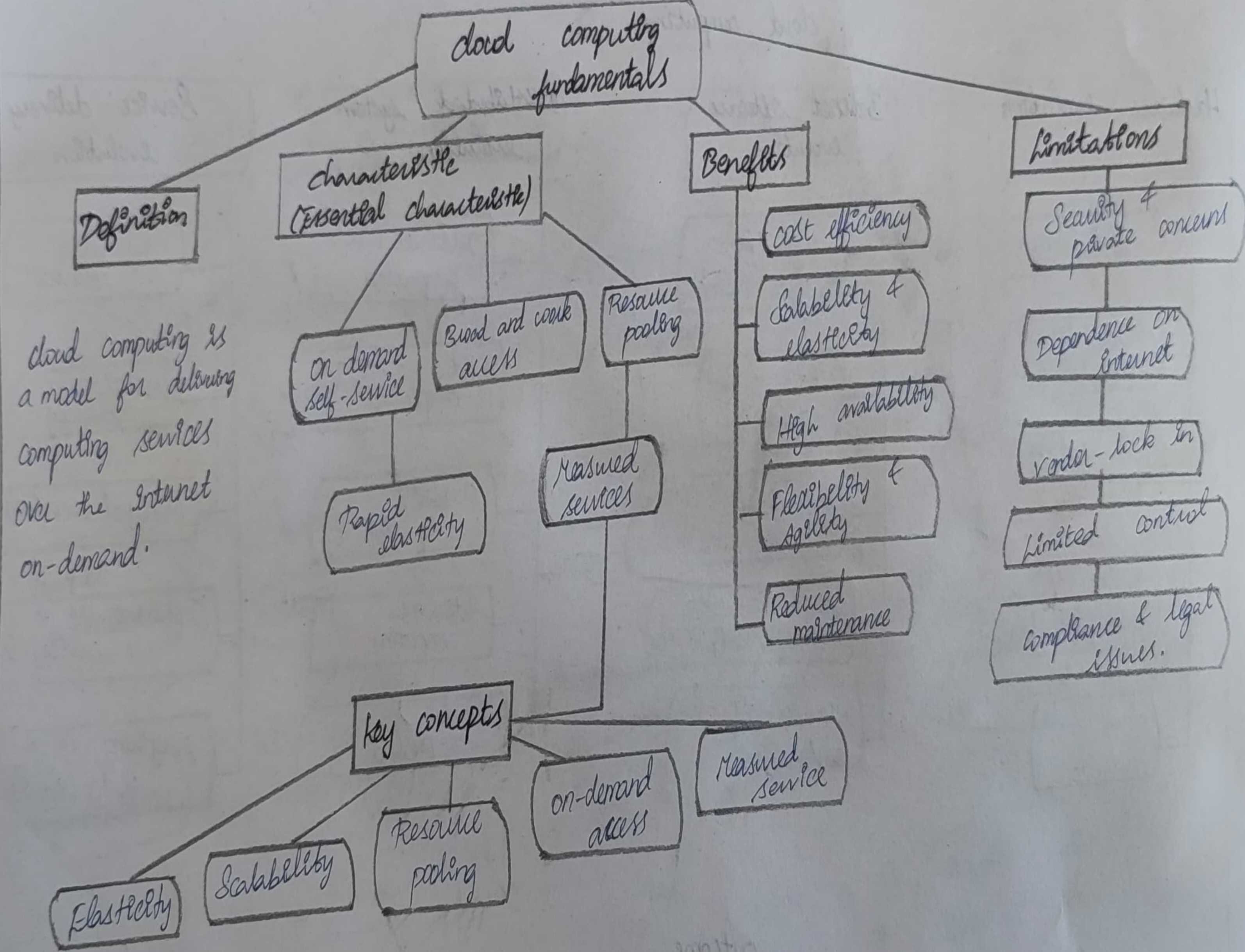
Assessment Tasks
Concept Mapping Tasks
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Concept Map 1 : cloud computing fundamentals



Concept map : 2 : Evolution of cloud computing

Evolution of cloud computing

- **Hardware evolution**
  - Main frames
  - client-server architecture
  - virtualization hardware
  - multi-core processor
  - cloud data centers.
- **Internet software evolution**
  - Web 1.0 (Static web)
  - Web 2.0 (Dynamic)
  - Web(2.0) (Semantic web)
  - APIs, SOA, web services (SOAP, REST)
  - Microservices & containers
- **Distributed system evolution**
  - Distributed computing
  - Grid computing
  - Utility computing
  - Autonomic computing
  - cloud computing
- **Service delivery evolution**
  - Physical resources
  - Hosted services
  - Managed services
  - Software as a service (SaaS)
  - Everything as a service (XaaS).

outcome

Scalable, Reliable, cost-effective, on-demand services delivered over the internet.

concept map 2 : Service models

cloud service models

Service model	IaaS (Infrastructure as a service)	PaaS (Platform as a service)	SaaS (Software as a service)
What it provides	virtualized computing resources over the Internet.	Platform and Tools for development & deployment.	complete Software application delivered over the Internet.
Provider responsibilities	Manage physical servers storage networking visualization.	Manage servers, storage networking runtime, middleware OS.	Managing everything including server, OS, applications and updates.
User responsibilities	Manage OS, middleware runtime, applications and data.	Develop test, display applications and manage data.	Use and configure applications ; manages user data.
Examples	AWS EC2, Microsoft Azure, VMS, Google compute engine.	Google app engine, Microsoft Azure, App service	Google workspace, Microsoft 365, Salesforce.

Concept map 4: Virtualization and Web services as enablers of cloud